



جامعة أم القرى
UMM AL-QURA UNIVERSITY

Big Data Mining – Final Project Report

Week 14

Google Maps Reviews Analysis Using Big Data Techniques

• Project Introduction

In our current era, we are witnessing rapid development across various technical fields, which has led to massive growth in the volume of data generated from numerous digital platforms and sources. Data has become ubiquitous, whether through social media platforms, e-service applications, or interactive websites, making the utilization of this vast amount of data crucial for extracting valuable information, patterns, and meaningful relationships to support and improve decisionmaking.

When dealing with massive and complex volumes of data, known as **Big Data**, it becomes essential to use tools and techniques capable of processing and analyzing this type of data with high efficiency. Big Data possesses specific characteristics unlike any other data type or structure; it differs from traditional data in terms of volume, variety, and velocity. This necessitates the adoption of specialized methodologies and frameworks that rely on high-capacity processing and analysis techniques to ensure accurate results and achieve better data quality.

The idea for this project stems from applying Big Data concepts and methodologies to textual data found on the **Google Maps** platform. More than one million records of reviews and comments—reflecting the opinions and experiences of visitors regarding tourist and entertainment destinations across various regions of the Kingdom of Saudi Arabia—have been collected. The project aims to analyze this textual data and extract sentiments and patterns associated with visitor experiences using data analysis and text mining techniques, contributing insights and indicators that can be utilized to enhance the quality of services and tourism experiences.



• Problem Statement

With the massive expansion in the use of electronic platforms, applications, and services, platforms such as Google Maps have come to provide enormous amounts of reviews and comments through which visitors share their experiences and opinions about tourist and entertainment destinations. This data represents a rich source of information that can be utilized to understand visitor satisfaction levels and analyze their experiences accurately.

Despite the great value of this data, dealing with and analyzing it poses a real challenge due to its massive volume, diversity, and unstructured textual nature. Relying on manual analysis for hundreds of thousands or millions of comments is impractical and consumes significant time and effort. Furthermore, the variation in writing styles and languages used, along with the presence of irrelevant or random data, increases the difficulty of extracting accurate and reliable results.

In addition, Arabic textual data faces numerous challenges related to Natural Language Processing (NLP), such as varying dialects, repetition, spelling errors, and the use of symbols and informal expressions. This makes the data cleaning and analysis process more complex compared to traditional structured data.

Accordingly, the project's problem lies in the need to build a methodology and workflow to analyze this massive textual data using Big Data and text mining techniques. The goal is to extract the sentiments and opinions of visitors regarding tourist and entertainment destinations in the Kingdom of Saudi Arabia, and to transform this data into information and insights that can be used to support decisionmaking and improve the quality of services and experiences provided to visitors.

• Project Objectives

This project aims to apply Big Data concepts and techniques alongside textual data analysis to understand and measure visitor opinions and experiences regarding tourist and entertainment destinations in the Kingdom of Saudi Arabia. This is achieved by collecting, processing, and analyzing a massive volume of textual reviews and comments extracted

from **Google Maps**. The specific objectives of the project are outlined in the following points:

- **Data Collection:** Gathering textual data from the Google Maps platform, including visitor reviews and comments regarding tourist and entertainment sites.
 - **Workflow Design:** Establishing a methodological plan and workflow for handling and processing Big Data using tools and techniques suited to the nature and size of the data.
 - **Data Cleaning and Preprocessing:** Performing cleaning and processing operations on textual data to improve data quality and remove non-useful elements such as symbols, duplicates, links, and random comments.
 - **Sentiment Analysis:** Applying sentiment analysis techniques to extract visitor sentiments and classify them into positive or negative categories.
 - **Aspect-Based Analysis:** Conducting analysis to discover patterns and trends within visitor opinions and extracting indicators related to their experiences at tourist and entertainment destinations.
 - **Data Visualization:** Presenting the results clearly through charts and interactive dashboards to support decision-making and enhance the quality of tourism and entertainment services.
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• Data Source Description

Reviews and comments published on the **Google Maps** platform are among the most prominent and important data sources for analyzing visitor opinions and experiences regarding tourist and entertainment destinations in the Kingdom of Saudi Arabia. This platform is characterized by being an interactive environment where users share their personal experiences, ratings, and various impressions of the places they have visited, making it a rich source of textual data ready for extraction and analysis.

Given the significant importance of this type of data, Google Maps was selected as the primary source for the project's data, due to the massive and diverse volume of textual reviews it provides that directly reflect visitor experiences. The **Apify** tool was utilized to collect the data and extract reviews and ratings from the platform, as it facilitated the aggregation processes and provided highly efficient access to the required data.

As previously mentioned, Apify was used in the data collection process from Google Maps, enabling the extraction of reviews and ratings for tourist and entertainment destinations in an organized manner suitable for analysis and processing. The data collection process involved several key steps:

- **URL Input:** Entering the specific Google Maps URL for the tourist or entertainment destination from which ratings and comments were to be extracted.
- **Defining Scope:** Determining the total number of ratings to be handled for each location through dedicated fields within the tool.
- **Volume Selection:** Choosing the specific number of comments to extract in proportion to the project's target data size.
- **Feature Selection:** Selecting the specific data columns (Features) to be collected after configuring the previous settings, as the tool provides a wide range of fields.
- **Data Export:** Saving the data after the collection process is complete in several formats, such as: **JSON, Excel, and CSV.**

• Structure and Nature Of Dataset

Column Name	Description
categoryName	Represents the type of tourist destination (e.g., Park, Farm, Forest, Museum, Archaeological Site, etc.).
city	The name of the city, governorate, or district where the location is situated.
location/lat & lng	Two separate columns representing the latitude and longitude coordinates for each tourist site.
neighborhood	The area or vicinity adjacent to the intended tourist landmark.
publishedAtDate	The date the review was written and published on Google Maps.
stars	The column containing the star ratings for the tourist destination.
street	The name of the street where the tourist site is located.
text	Represents the written comments and reviews provided by users who visited the location.
title	The actual name and title of the tourist landmark.

The work was distributed among team members according to the 13 administrative regions of the Kingdom of Saudi Arabia. To organize the collection process more efficiently, these regions were categorized into five main geographical areas. Each member was assigned to a specific area encompassing its most prominent tourist and entertainment destinations, with the objective of gathering the largest possible volume of reviews and comments related to those locations.

Region	Included Areas
Central Region	Riyadh, Qassim
Western Region	Makkah, Madinah
Eastern Region	Dammam, Al-Ahsa, Al-Khobar, and other cities
Southern Region	Al-Baha, Asir, Jazan, Najran
Northern Region	Hail, Tabuk, Al-Jouf, Northern Borders

• Overview of the Big Data Ecosystem

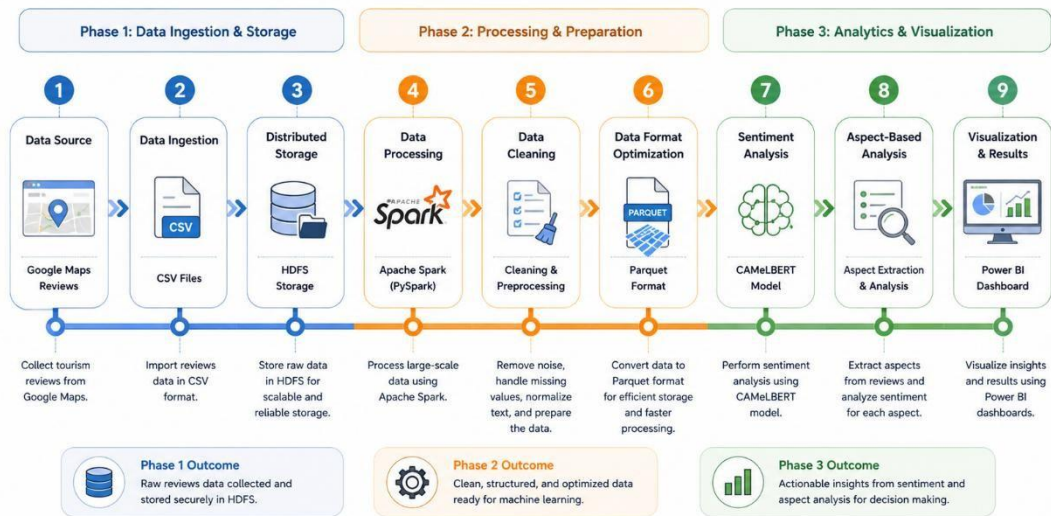
The system was designed to handle Big Data through multiple interconnected stages for processing large-scale textual review data collected from Google Maps. The ecosystem begins with the data source layer, followed by data ingestion and distributed storage, then the processing and analytics stages, and finally the visualization and output layer.

Different technologies were utilized within the system according to the role of each stage. HDFS was used for distributed storage, while Apache Spark was utilized for large-scale data processing with high efficiency. In addition, the Parquet format was adopted to improve storage efficiency and processing performance. Machine learning techniques were also applied to perform sentiment analysis and aspect-based analysis. Finally, Power BI was used to visualize results and analytical insights.

The overall environment was designed to improve processing efficiency and provide the ability to handle large volumes of textual data within an integrated Big Data framework.

Big Data Pipeline for Tourism Reviews Analysis

• 9-Step System Workflow •



The figure shows Big Data Pipeline for Tourism Reviews Analysis

• Data Ingestion

The data ingestion stage represents the process of integrating the collected dataset into the Big Data environment in preparation for storage and processing stages. In this project, the review files collected from Google Maps were organized in a structured folder hierarchy and handled in CSV format before being transferred into the distributed environment.

This stage focused on preparing the raw dataset for large-scale processing by ensuring that the files were properly organized and accessible for later stages within the Big Data ecosystem.

• Distributed Storage

The **HDFS** (Hadoop Distributed File System) was utilized as the distributed storage layer within the Big Data environment, aiming to store and manage large-scale data with high efficiency. Using HDFS helped provide a storage environment capable of handling the massive volume of textual data, while supporting scalability and file management within a distributed system.



Furthermore, this stage contributed to improving data accessibility and preparing the data for processing and analysis operations within the Big Data ecosystem.

• Data Processing

PySpark was utilized during this stage to process data within the Big Data environment, leveraging its high capacity for large-scale data processing through distributed computing. This phase involved reading data files, handling the distributed data structure, and preparing the data for subsequent analysis operations.

This stage contributed to enhancing processing efficiency and handling the massive volume of textual data more effectively within the Big Data ecosystem.

• Data Cleaning

The data cleaning and initial preprocessing phase was previously executed as part of the graduation project, encompassing several operations aimed at enhancing data quality and preparing it for analysis and modeling. This stage included text processing, data organization, and addressing issues specifically associated with Arabic textual data.

In this project, the pre-processed data was utilized while employing Big Data techniques to handle the data within a distributed environment, ensuring it was ready for the subsequent stages of analysis and processing.

The data cleaning and preprocessing operations included several stages aimed at improving the quality of the textual data and preparing it for analysis and modeling. These stages involved standardizing column names, handling missing values, and cleaning texts by removing links and unnecessary symbols. Additionally, Arabic text normalization was applied, along with addressing character repetition. The texts were also prepared to be compatible with various machine learning models, alongside performing data quality validation and analyzing emojis associated with sentiment.

• Data Format Optimization

During this stage, the data storage format was optimized to enhance processing efficiency within the Big Data environment, converting the data from CSV files to the **Parquet** format. Utilizing the Parquet format contributed to reducing storage size and improving data reading and processing speeds, in addition to providing better performance when handling Big Data within Apache Spark .

• Sentiment Analysis

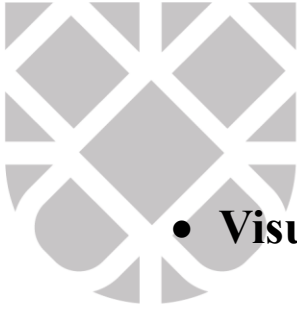
In this stage, sentiment analysis was applied to the textual reviews to classify visitor opinions into positive or negative sentiments. The **CAMeLBERT** model was utilized due to its high capability in handling Arabic texts and understanding the various dialects and writing styles used within the reviews.

This phase contributed to transforming unstructured textual data into analyzable data, enabling the processing of large-scale reviews and the extraction of results related to sentiment trends within the data.

• Aspect-Based Analysis

In this stage, Aspect-Based Analysis was applied to extract the most frequent themes within the textual reviews and analyze visitor opinions related to each aspect separately. A **Rule-Based** approach was adopted using a **Dictionary** containing a set of keywords associated with each aspect—such as cleanliness, service, prices, crowding, and other factors relevant to tourist and entertainment destinations.

Once the aspects were identified within the texts, the **CAMeLBERT** model was used to analyze the sentiment associated with each specific aspect. This enabled the extraction of more detailed indicators regarding visitor opinions and satisfaction levels tied to each individual aspect within the textual data.



• Visualization & Results Discussion

• Results

Dataset	Accuracy	Negative Class			Positive Class		
		Precision	Recall	F1-Score	Precision	Recall	F1-Score
Google Maps	93%	81%	45%	58%	94%	99%	96%

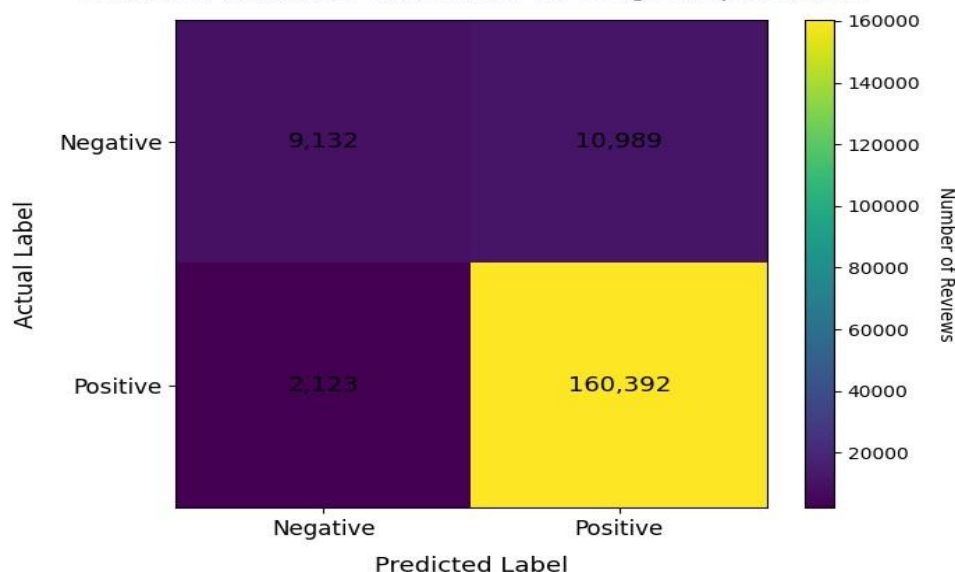
The **CAMeLBERT** model achieved strong performance on the Google Maps data, with an overall accuracy reaching **93%**. The model demonstrated high performance in classifying positive reviews, achieving a **Precision** of **94%**, a **Recall** of **99%**, and an **F1-Score** of **96%**. These results indicate the model's high capability in understanding positive sentiments within Arabic textual reviews.

In contrast, the negative class performed lower compared to the positive class, with a

Precision of **81%**, a **Recall** of **45%**, and an **F1-Score** of **58%**. The decrease in the **Recall** value suggests that a portion of negative reviews were misclassified as positive. This is largely attributed to data imbalance, as positive reviews represent the majority of the data records.

Overall, the results demonstrate the ability of the **CAMeLBERT** model to handle large-scale Arabic textual data and extract sentiments with high efficiency, particularly in classifying positive sentiments within tourism and entertainment reviews.

Confusion Matrix for CAMELBERT on Google Maps Dataset

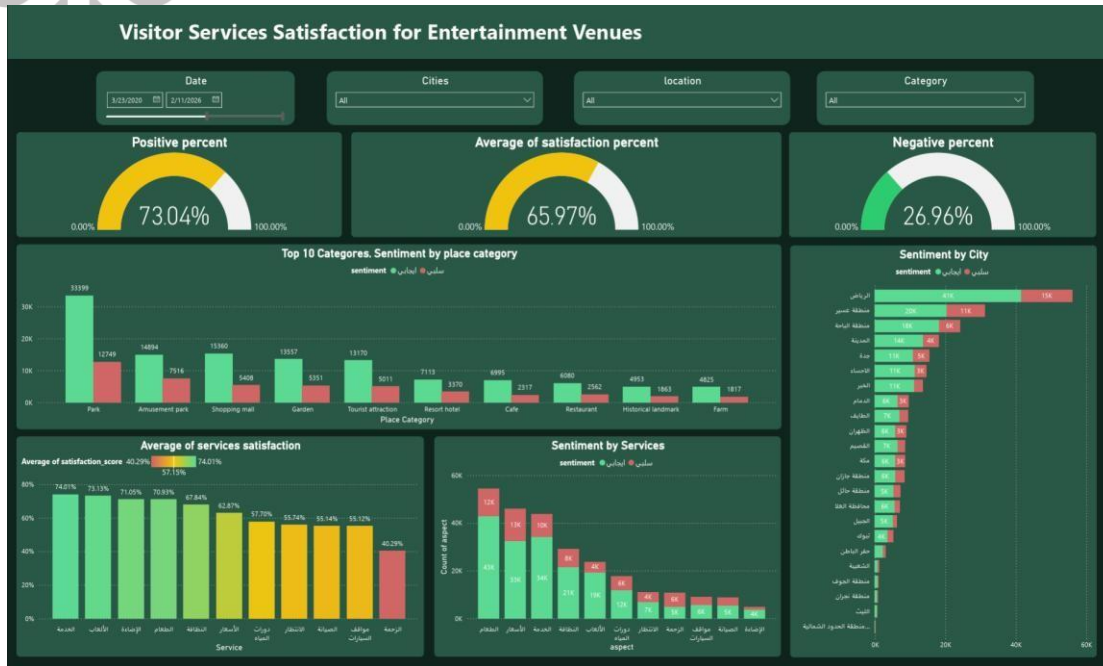


Confusion Matrix Show the performance of the **CAMELBERT** model in classifying positive and negative reviews on the Google Maps dataset. It is observed that the model successfully classified a large number of positive reviews correctly, with **160,392** positive reviews accurately identified, compared to a relatively small number of positive reviews misclassified as negative.

In contrast, the model showed lower performance in classifying negative reviews, with **9,132** negative reviews correctly identified, while **10,989** negative reviews were misclassified as positive. This reflects a greater difficulty in detecting some negative reviews compared to positive ones.

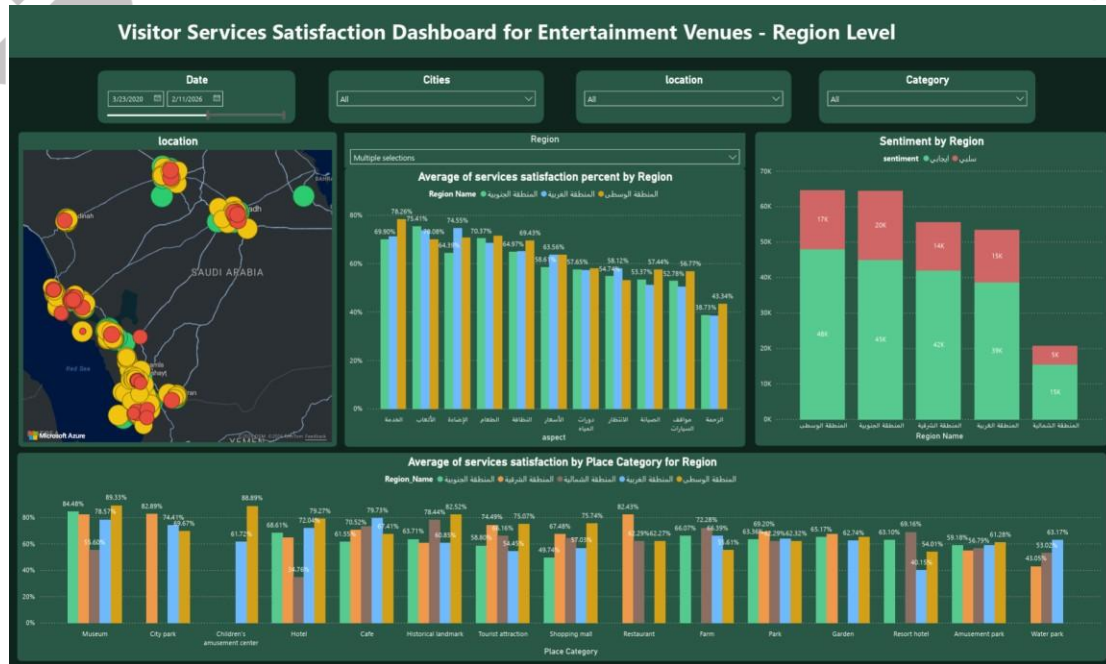
These results indicate that while the model possesses a high capability for understanding and classifying positive sentiments within Arabic textual data, the data imbalance between the two classes has impacted the model's performance in accurately classifying negative reviews.

• Dashboards and Analytics



The interactive dashboard shows sentiment analysis indicators and visitor satisfaction levels for entertainment venues based on Google Maps reviews. The dashboard includes a set of interactive filters—such as date, city, location, and category—allowing for flexible data exploration and analysis.

The dashboard shows that the percentage of positive sentiment reached **73.04%** compared to **26.96%** for negative sentiment, while the average overall satisfaction level reached **65.97%**. Additionally, the dashboard presents a comparison between different venue categories in terms of the number of positive and negative reviews; the "Parks" category was among the highest in receiving positive feedback, with over **33,000** positive reviews. The dashboard also shows the distribution of sentiment by city, where Riyadh recorded the highest number of positive reviews with over **41,000**. Furthermore, the dashboard includes an aspect-based analysis of various service factors such as cleanliness, prices, service, crowding, and parking. This helps identify the most influential aspects of visitor satisfaction and provides insights to support improving the quality of tourism and entertainment services.



The interactive dashboard shows an analysis of visitor satisfaction levels for entertainment venues across the administrative regions of Saudi Arabia, based on Google Maps reviews and sentiment analysis of textual data. The dashboard includes a set of interactive filters such as date, city, location, and category, in addition to the ability to filter by region.

The dashboard shows the distribution of positive and negative sentiments across the Kingdom's regions, where the **Central Region** and the **Southern Region** recorded the highest number of positive reviews, with approximately over **45,000** positive reviews for each region. In contrast, the **Northern Region** recorded the lowest number of reviews compared to other regions.

Furthermore, the dashboard displays the average visitor satisfaction level based on various service aspects—such as service, games/rides, cleanliness, prices, crowding, and parking—with comparisons between different regions. Additionally, it features an interactive map showing the geographical distribution of entertainment venues within the Kingdom, along with a satisfaction level comparison by venue category, such as museums, parks, hotels, restaurants, and entertainment centers.

These analyses help provide a clearer vision regarding the variation in visitor satisfaction across Saudi regions and identify the aspects and services that require improvement to support the quality of tourism and entertainment services.



The interactive dashboard shows an analysis of visitor satisfaction levels for entertainment venues across cities in the Kingdom of Saudi Arabia, based on sentiment analysis results of textual reviews extracted from Google Maps. The dashboard includes a set of interactive filters—such as date, city, location, category, and service—allowing for flexible and detailed data analysis.

The dashboard shows the evolution of the average visitor satisfaction level over the years; the year **2024** recorded the highest average satisfaction at **68.16%**, while the average decreased in **2026** to **56.61%**. Additionally, the dashboard presents a comparison between cities in terms of average satisfaction levels; **Al Jubail** ranked first with a satisfaction rate of **74.71%**, followed by **Al Ula** at **74.27%**, and then **Al Khobar** at **74.24%**.

Furthermore, the dashboard includes an interactive comparison between cities over the years to measure changes in satisfaction levels over time, which helps in tracking the performance of different cities and analyzing visitor satisfaction trends. These analyses contribute to providing clearer insights into the cities achieving the highest visitor satisfaction, as well as identifying areas that may require improvements in the quality of tourism and entertainment services.

• Tool-Layer Justification

Tool / Technology	Purpose of Use
Apify	Used for collecting review data from Google Maps efficiently and quickly.
Google Maps	Used as the primary source of textual reviews and visitor feedback.
Google Colab	Used during the data processing stage and file reading operations.
HDFS	Used as a distributed storage system for handling large-scale data.
PySpark	Used for efficient Big Data processing through distributed computing.
Parquet	Used to improve storage efficiency and accelerate data reading and processing.
Kaggle	Used to provide a suitable execution environment and GPU support for running the CAMELBERT model.
CAMELBERT	Used for sentiment analysis due to its ability to understand Arabic text and different dialects.
Power BI	Used for building interactive dashboards and visualizing results and analytics.

• Big Data Justification

Why is this project a Big Data solution rather than a traditional data analysis task?

This project is considered a big data project due to its reliance on large-scale textual data containing more than one million reviews collected from Google Maps. Handling this large volume of data requires techniques capable of high-efficiency storage and processing, which exceeds the capabilities of traditional analysis methods.

The project utilized an integrated big data environment that included HDFS for distributed storage and PySpark for distributed processing, in addition to using the Parquet format to improve storage and processing efficiency. Machine learning techniques were also relied upon to analyze large-scale



Arabic textual data and extract sentiments and various aspects from the reviews.

- **Ethics and Privacy**

In this project, ethical and privacy aspects were adhered to during data handling, as publicly available data from Google Maps was used without collecting any sensitive personal information regarding the users. Additionally, the focus was placed on using the data for academic and analytical purposes only.

- **Use of AI Tools**

AI tools were used in a limited capacity to support some auxiliary aspects of the project, such as assisting in writing some programming codes, content paraphrasing processes, improving grammatical rules, and translation processes between Arabic and English. AI tools were also utilized in formatting some parts of the content and assisting in the design of the Big Data Ecosystem diagram.

However, the stages of processing, data analysis, programmatic modeling, and building results and dashboards were actually executed by the project team.

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